

ORIGINAL ARTICLE

Validation of Hindi version of the Pittsburgh Sleep Quality Index

Anil Kumar^{1*}, Rohini Handa², Sundeep Kumar Upadhyaya², Sirinder Jit Gupta²**Abstract**

Background: Sleep quality is poorly understood in the Indian population. Lack of a suitable tool to assess sleep quality in Hindi language is the main reason behind this. Pittsburgh sleep quality index (PSQI) is widely used in different population groups to assess subjective sleep quality over last one month.

Objectives: To assess the reliability and validity of the Hindi translated version of Pittsburgh Sleep Quality Index (PSQI-H).

Methods: The PSQI-H was developed from PSQI according to following steps: (a) translation, (b) back-translation, (c) comparison between translation and back-translation performed by a group of experts, and (d) pre-pilot test in intended population. The PSQI-H was applied to 105 bilingual individuals knowing Hindi and English. The internal consistency of PSQI-H was assessed by Cronbach's alpha. For test-retest reliability assessment intraclass correlation coefficient (ICC) was measured between PSQI-H at baseline and PSQI after 4 weeks. The Pearson's coefficient was used to assess the correlation between the score of the questions and the PSQI-H scores.

Results: The seven components of PSQI-H shows acceptable level of internal consistency with Cronbach's alpha of 0.776. There is good test-retest reliability between PSQI-H and PSQI as measured by ICC of 0.979. The score of individual items and global scores of PSQI-H were highly correlated with each other ($p < 0.001$). The mean of the seven individual components score and global scores of PSQI-H at baseline and original PSQI after 4 weeks did not differ significantly.

Conclusion: This study results demonstrate that the PSQI-H is a valid and reliable instrument for the assessment of sleep quality. PSQI-H can be used for the assessment of sleep quality in the predominantly Hindi speaking population.

Introduction

Sleep quality is a major domain of the health-related quality of life (HRQoL). Poor sleep quality and/or disorders have been found to be associated with somatic pain, fatigue, anxiety, stress, depression, decreased attention, drowsy driving, risk-taking behavior, impaired social relationships, reduced cognitive performance, altered sympathetic activity, increased risk of cardiovascular events, and overall poor health.^{1,2} Polysomnography (PSG) is considered the gold standard for sleep quality assessment and sleep disorder diagnosis. But the use of PSG in developing countries is limited by its cost, limited availability, time-consuming nature and lack of trained personnel. For these reasons, self-report retrospective and prospective questionnaires have been developed to

assess sleep quality in both normal and diseased individuals. These subjective measurement tools can be used in clinical practice and in epidemiological studies for diagnostic purposes and monitoring treatment responses.^{3,4} The Pittsburgh Sleep Quality Index (PSQI) is a self-administered questionnaire to assess subjective sleep quality over the last one month. The PSQI has good reliability and validity and demonstrates good sensitivity and specificity to distinguish good and poor sleepers.⁵ The PSQI has been validated and translated into several languages. The PSQI has also been used and validated in several different patient populations, such as patients with psychiatric or sleep disorders, kidney, digestive, or respiratory diseases, cancer, and rheumatic disease patients.⁶⁻¹¹ India is the second most populous country in the world. Hindi

is the most widely spoken language in India and enjoys the distinction of being the 4th most widely spoken language in the world. A validated version of PSQI is not available in the Hindi language. A Hindi translation and validation of the PSQI would provide an internationally standardized questionnaire for sleep quality assessment for Hindi speaking population not only in India but also widely scattered Indian diaspora all over the world. The aim of this study was to translate the PSQI into Hindi and to assess its validity, and reliability in a sample of the Indian population.

Materials and Methods**Type of Study**

This was a longitudinal prospective study conducted at a tertiary care hospital at New Delhi, India.

Study Participants

A total of 105 bilingual individuals knowing Hindi and English, participated in this study. Study participants were healthy hospital staff members of age 18-65 years. Individuals with complaints of confounding medical conditions such as psychiatric illness, obstructive sleep apnea, chronic pain, recent surgery, current or past history of drug abuse and those who were doing night duty were excluded from the study. Participants were briefed on the purposes of the research, and all gave their written informed consent. The study was approved by research ethics committee.

Research Tool

PSQI measures the patient's self-reported sleep quality over the last month. PSQI consists of 19 items and 5 additional questions. The latter five questions are not included in scoring and rated by bed partner or roommate. These 19 items are combined to form 7

¹Resident, ²Consultant, Department of Rheumatology, Indraprastha Apollo Hospital, New Delhi; *Corresponding Author

Received: 20.12.2020; Accepted: 19.01.2021

Table 1: Cronbach's Alpha of individual items of PSQI-H

Components	Scale Mean if Item Deleted	Scale Variance if Item Deleted	Cronbach's Alpha if Item Deleted
Sleep Quality	4.10	5.741	.688
Sleep Latency	3.79	5.783	.754
Sleep Duration	4.70	7.060	.751
Sleep Efficiency	4.90	6.952	.737
Sleep Disturbances	4.18	8.227	.778
Use Of Sleep Medicines	5.08	7.956	.775
Daytime Dysfunction	4.46	6.539	.736

Table 2: Intraclass correlation coefficient (ICC) between PSQI-H at baseline and PSQI after 4 weeks

Component	ICC	95% CI	P-value
Sleep quality	0.939	0.910-0.958	p<0.001
Sleep latency	0.946	0.921-0.964	p<0.001
Sleep duration	0.975	0.963-0.983	p<0.001
Sleep efficiency	0.952	0.929-0.967	p<0.001
Sleep disturbance	0.906	0.862-0.936	p<0.001
Sleep medicine uses	0.957	0.936-0.971	p<0.001
Daytime dysfunction	0.883	0.828-0.921	p<0.001
Global PSQI	0.979	0.97-0.986	p<0.001

components of sleep quality: subjective sleep quality, sleep latency, sleep duration, habitual sleep efficiency, sleep disturbance, sleep medication use, and daytime dysfunction. Each component has a score that ranges from 0 to 3. A global PSQI score is obtained by summing the 7 component scores (range = 0-21). Participants were dichotomized into a poor-sleep quality group if the PSQI was ≥ 5.0 and a good-sleep quality group if the PSQI was < 5.0 .⁵

Translation

The PSQI was translated into Hindi by 2 independent bilingual translators. After discussion among the experts, translators, and researcher, a Hindi translated PSQI was obtained. Back-translations into English were carried out by another 2 independent translators. The comparison of the PSQI with the back-translation was made by research committee members who were fluent in English and Hindi. After detail analysis of each item, the pre-final version of the PSQI in Hindi was obtained. A pre-pilot test was done to assess the linguistic interchangeability of the prefinal version of the translated PSQI questionnaire on a sample of 25 respondents and then original PSQI after 2-week intervals. After completing

Table 3: Pearson's correlation coefficient between seven components and global score of PSQI-H

Components	Pearson's Correlation(r)	P-value
Sleep Quality	0.856	p<0.001
Sleep Latency	0.736	p<0.001
Sleep Duration	0.627	p<0.001
Sleep Efficiency	0.688	p<0.001
Sleep Disturbances	0.431	p<0.001
Use Of Sleep Medicines	0.474	p<0.001
Daytime Dysfunction	0.704	p<0.001

the translated questionnaire, the respondent was asked to elaborate what they thought each questionnaire item and their corresponding response meant. The scores of each version were correlated with each other. After incorporating inputs from study participants a final version of Hindi translated PSQI (PSQI-H) was made.

Procedures

The PSQI-H was applied to 105 study participants in Indraprastha Apollo Hospital, New Delhi, between September 2018 and December 2018. After 4 weeks of administration of the PSQI-H, 105 study subjects were administered original PSQI (PSQI-E) to assess test-retest reliability. All volunteers were given general instructions for filling the PSQI by a simple demonstration.

Statistical Analysis

Data were analyzed using the SPSS version 20 package for Windows. Quantitative variables were expressed as mean $\pm$ SD, and qualitative variables were expressed as percentage values. Two-tailed P values of 0.05 or less were regarded as statistically significant, and 95% confidence intervals were calculated for results. The internal consistency of the PSQI-H was assessed by calculating Cronbach's alpha coefficients using the seven component scores of the PSQI. Cronbach's $\alpha = 0$ indicates no internal consistency, whereas $\alpha = 1$ reflects perfect internal consistency. Cronbach's alpha of at least 0.70 suggests adequate internal consistency. Test-retest reliability was measured by the intra-class correlation coefficient (ICC).¹² Scores on individual items of the scale at baseline were compared with respective scores after 4 weeks by t-test. Pearson's coefficient was used to assess the correlation between the scores of the question and the PSQI-H scores.

Table 4: Comparison between scores for individual items of PSQI-H at baseline and original PSQI after 4 weeks

Component	PSQI-H	PSQI-E	P value
Sleep Quality	1.1 $\pm$ 0.759	1.17 $\pm$ 0.686	0.052
Sleep Latency	1.42 $\pm$ 0.938	1.35 $\pm$ 0.888	0.109
Sleep Duration	0.5 $\pm$ 0.622	0.49 $\pm$ 0.622	0.32
Sleep Efficiency	0.3 $\pm$ 0.587	0.32 $\pm$ 0.612	0.259
Sleep Disturbance	1.02 $\pm$ 0.366	1.05 $\pm$ 0.377	0.181
Sleep Medicine Uses	0.12 $\pm$ 0.454	0.11 $\pm$ 0.375	0.566
Daytime Dysfunction	0.74 $\pm$ 0.707	0.76 $\pm$ 0.714	0.672
Global PSQI	5.2 $\pm$ 3.017	5.26 $\pm$ 2.981	0.494

Results

The sample consisted of 105 participants. The mean age of participants was 35.3 ± 7.5 years, 42.8% (45) participants were male and 57.2% (60) participants were female. PSQI-H global score ranged from 1-14. The mean PSQI-H global score for the total sample was 5.2 ± 3.0 . Using the recommended cut-off ≥ 5.0 , 48 (45.7%) of the participants reported poor sleep quality. The internal consistency analysis of PSQI-H yielded a Cronbach's alpha of 0.776, which is an acceptable value for the reliability of the questionnaire. The internal consistency did not change with the removal of specific items except for sleep quality (Table 1). Intraclass correlation coefficient (ICC) was used for test-retest reliability assessment, between PSQI-H and original PSQI. Our results showed statistically significant correlations between the global score and individual components of the PSQI-H at baseline and PSQI after 4 weeks. The intraclass correlation coefficient ranged from 0.883 to 0.975 for the seven components and reached 0.979 for the global score (Table 2). Pearson's correlation coefficients were calculated among the seven components and the global score of PSQI-H. The correlations between the global score and seven components of PSQI-H was statistically significant ($p < 0.001$), but the correlation between sleep disturbances and the global score was the lowest (Table 3). We found that the mean of the global score of Hindi translated version of PSQI at baseline and original PSQI after 4 weeks did not differ significantly (5.2 ± 3.017 and 5.26 ± 2.981 , $p = 0.494$). The mean of individual items of PSQI-H and PSQI were calculated and compared. As shown in Table 4, there was no

पिट्सबर्ग निद्रा गुणवत्ता सूचकांक (PSQI)

रोगी का नाम क्रमांक संख्या तारीख समय

निर्देश:

निम्नलिखित प्रश्न केवल पिछले महीने के दौरान आपकी नींद की सामान्य आदतों के बारे में हैं। आपके उत्तर पिछले एक महीने के अधिकांश दिनों और रातों को सबसे सटीक रूप में दर्शाने वाले होने चाहिए। कृपया सभी सवालों का जवाब दें।

पिछले महीने के दौरान,

1. आप आमतौर पर रात को किस समय बिस्तर में सोने के लिए जाते थे?
2. आपको आमतौर पर हर रात बिस्तर में लेटने के बाद सोने में कितनी देर (मिनट) लगती थी?.....
3. आप आमतौर पर सुबह को नींद से कितने बजे जागते थे?.....
4. आप रात में वास्तव में कितने घंटे सोते थे?..... (यह समय बिस्तर में बिताए गए समय से अलग हो सकता है।)

5. पिछले महीने के दौरान, आपको कितनी बार ठीक से सोने में परेशानी हुई है। क्योंकि	पिछले महीने के दौरान नहीं	हफ्ते में एक बार से कम	हफ्ते में एक या दो बार	हफ्ते में तीन बार या ज्यादा
क) आपको 30 मिनट के अन्दर नींद नहीं आई				
ख) आपकी नींद बीच रात में या सुबह जल्दी खुल गई				
ग) आपको लघुशंका के बिस्तर से उठना पड़ा				
घ) आप आसानी से सांस नहीं ले पा रहे थे				
ङ.) आपको खौसी थी या आप जोर से खरटे ले रहे थे				
च) आपको बहुत ज्यादा ठंड लग रही थी				
छ) आपको बहुत ज्यादा गर्मी लग रही थी				
ज) आपको बुरे सपने आते थे				
झ) आपको दर्द था				
ण) कुछ दूसरे कारण थे : कृपया वर्णन करें, पिछले महीने के दौरान, आपको कितनी बार इसकी वजह से सोने में परेशानी हुई				

6. पिछले महीने के दौरान, आपकी नींद की गुणवत्ता का मूल्यांकन आपके अनुसार कैसा रहा	बहुत अच्छा	कुछ हद तक अच्छा	कुछ हद तक खराब	बहुत खराब
--	------------	-----------------	----------------	-----------

7. पिछले महीने के दौरान, आपने सोने के लिए कितनी बार दवाओं नींद की दवाओं का सेवन किया था	पिछले महीने के दौरान नहीं	हफ्ते में एक बार से कम	हफ्ते में एक या दो बार	हफ्ते में तीन बार या ज्यादा
---	---------------------------	------------------------	------------------------	-----------------------------

8. पिछले महीने के दौरान, वाहन चलाते हुए, भोजन करते हुए, या कोई सामाजिक गतिविधि करते हुए, आपको कितनी बार नींद महसूस हुई थी	पिछले महीने के दौरान नहीं	हफ्ते में एक बार से कम	हफ्ते में एक या दो बार	हफ्ते में तीन बार या ज्यादा
---	---------------------------	------------------------	------------------------	-----------------------------

9. पिछले महीने के दौरान, अपने कार्यों को पूरा करने के लिए पर्याप्त उत्साह बनाए रखना आपके लिए कितना मुश्किल था	बिल्कुल मुश्किल नहीं था	सिर्फ थोड़ा सा मुश्किल था	काफी मुश्किल था	बहुत ज्यादा मुश्किल था
---	-------------------------	---------------------------	-----------------	------------------------

Scoring methods available at www.sleep.pitt.edu/instruments

significant difference between the mean score of individual items of both the scales (p-value >0.05 for all items).

Discussion

There is a relative paucity of studies on sleep quality in India. Non-availability of any suitable and

validated questionnaires in local languages is the main reason behind this. Some studies in India have used the original English PSQI, however, the majority of the Indian population does not understand English.¹³⁻¹⁶ Hindi is most commonly spoken and national language of India. This study translates

the PSQI into Hindi and validates it. The present study shows that the PSQI Hindi is a reliable instrument, with a good convergent and discriminant construct validity, for measuring the subjective perception of sleep quality and patterns in Hindi speaking population. Use of questionnaire in a

group of bilingual individuals shows good linguistic interchangeability between the PSQI-H and PSQI. The items in the Hindi version were easily understood and completed in 8-10 minutes. PSQI-H had an acceptable internal consistency with Cronbach's α of 0.776. The subscale scores were found to be satisfactory for validity when compared with the original English questionnaire. The results were similar to the original PSQI questionnaire, which was tested in Indian population.¹⁷ In Bangla speaking Indian population the Bengali version of PSQI shows good internal consistency with Cronbach's α of 0.816.¹⁸ Additionally, the results were similar to the previous studies that tested the internal consistency of the PSQI in other languages.¹⁹⁻²¹ There is a good test-retest reliability agreement between the original PSQI and PSQI-H. The test-retest reliability of the PSQI-H global score over 4 weeks interval showed an intraclass correlation coefficient (ICC) of 0.979 and ranged from 0.883 to 0.975 for the seven components. The mean score of individual items and the global score of PSQI-H were similar to PSQI after 4-week interval, indicating good linguistic interchangeability. The global score and individual components of PSQI-H were strongly correlated with each other. Our study has several advantages. We assessed the PSQI in the Hindi language as it is 4th most widely spoken language in the world. To the best of our knowledge, this is the first validated Hindi version of PSQI. The internal consistency of the PSQI-H was in an acceptable range and similar to previously reported data in the Indian population. There was a strong correlation between English and Hindi PSQI as well as between individual questions. This study has

some limitations. Only perceived sleep quality was evaluated by questionnaires, sleep quality was not tested by gold standard polysomnography. The study sample comprised normal hospital staff members. Additional studies require study participants with medical conditions which affects sleep. In this study, the sample size was small. A larger sample size would have been ideal but was not possible due to logistic reasons.

Conclusion

The Hindi version of PSQI showed good internal consistency and validity. The Hindi version of the PSQI can be used for the assessment of sleep quality in the predominantly Hindi speaking population.

Acknowledgments

The authors thank Dr. Buysse for giving permission to translate and use the PSQI in the present study. We gratefully acknowledge Dr. Shweta Gupta (MBBS, DMRD), Mr. Satpal Saini (MA, BEd, MPhil), Mr. Radhey Shyam (MA, MPhil) and Dr. Harish Dahiya (MA, MPhil, Ph.D.) for forward translation and backward translation of questionnaires. We sincerely thanks all the participants who took part in this study.

References

- Gaultney JF. The prevalence of sleep disorders in college students: impact on academic performance. *J Am Coll Health* 2010; 59:91-97.
- Sweilheh WM, Ali IA, Sawalha AF, et al. Sleep habits and sleep problems among Palestinian students. *Child Adolesc Psychiatry Ment Health* 2011; 5:25.
- Weaver TE. Outcome measurement in sleep medicine practice and research. Part 1: assessment of symptoms, subjective and objective daytime sleepiness, health-related quality of life and functional status. *Sleep Med Rev* 2001; 5:103-128.
- Krystal AD, Edinger JD. Measuring sleep quality. *Sleep Med* 2008; 9:510-517.
- Buysse DJ, Reynolds CF 3rd, Monk TH, et al. The Pittsburgh Sleep Quality Index: a new instrument for psychiatric practice and research. *Psychiatry Res* 1989; 28:193-213.
- Guo S, Sun W, Liu C, et al. Structural Validity of the Pittsburgh Sleep Quality Index in Chinese Undergraduate Students. *Front Psychol* 2016; 7:1126.
- Sekercioglu N, Curtis B, Murphy S, et al. Sleep quality and its correlates in patients with chronic kidney disease: a cross-sectional design. *Ren Fail* 2015; 37:757-762.
- Xiao G, Ye Q, Han T, et al. Study of the sleep quality and psychological state of patients with hepatitis B liver cirrhosis. *Hepatol Res* 2018; 48:E275-E282.
- De S. Subjective assessment of quality of sleep in chronic obstructive pulmonary disease patient and its relationship with associated depression. *Lung India* 2012; 29:332-335.
- Akman T, Yavuzsen T, Sevgen Z, et al. Evaluation of sleep disorders in cancer patients based on Pittsburgh Sleep Quality Index. *Eur J Cancer Care* 2015; 24:553-559.
- Sariyildiz MA, Batmaz I, Bozkurt M, et al. Sleep quality in rheumatoid arthritis: relationship between the disease severity, depression, functional status and the quality of life. *J Clin Med Res* 2014; 6:44-52.
- Tsang S, Royse CF, Terkawi AS. Guidelines for developing, translating, and validating a questionnaire in perioperative and pain medicine. *Saudi J Anaesth* 2017; 11:S80-S89.
- Bhardwaj M, Arumugam N, Gambhir S. Efficacy of cranial electrical stimulation and rational emotive behavior therapy in improving psychological illness among chronic stroke survivors: A pilot randomized controlled trial. *Ann Indian Acad Neurol* 2018; 21:188-192.
- Panda S, Taly AB, Sinha S, et al. Sleep-related disorders among a healthy population in South India. *Neurol India* 2012; 60:68-74.
- George S, Paul G, Paul N. Study on sleep quality and associated psychosocial factors among elderly in a rural population of Kerala, India. *Int J Community Med Public Health* 2018; 5:526-531.
- Pattanaik A, Priyadarshini A, Pattanaik N. Relation between sleep quality and circulatory levels of inflammatory marker in women of reproductive age group. *Natl J Physiol Pharm Pharmacol* 2018; 8:1414-1417.
- Manzar MD, Moiz JA, Zannat W, et al. Validity of the Pittsburgh Sleep Quality Index in Indian University Students. *Oman Med J* 2015; 30:193-202.
- Mondal H, Mondal S, Baidya C. Comparison of perceived sleep quality among urban and rural adult population by Bengali Pittsburgh sleep quality index. *Adv Human Biol* 2018; 8:36-40.
- Hita-Contreras F, Martínez-López E, Latorre-Román PA, et al. Reliability and validity of the Spanish version of the Pittsburgh Sleep Quality Index (PSQI) in patients with fibromyalgia. *Rheumatol Int* 2014; 34:929-936.
- Bertolazi AN, Fagundes SC, Hoff LS, et al. Validation of the Brazilian Portuguese version of the Pittsburgh Sleep Quality Index. *Sleep Med* 2011; 12:70-75.
- Suleiman KH, Yates BC, Berger AM, et al. Translating the Pittsburgh Sleep Quality Index into Arabic. *West J Nurs Res* 2010; 32:250-268.